

TensorEngine: informe de corrida

Modelo: draft4_case_2

Run ID: run_8a58954981b2776b7ff8

Estado: partial

Backend: structural-python 0.9.0

Verificaciones: 43 aprobadas, 0 fallidas, 2 indeterminadas.

Presentación: simplificación protegida, sin modificar los objetos canónicos. Las operaciones e hipótesis usadas por expresión constan en `presentation.json` y en los comentarios del archivo LaTeX. La validación xAct corresponde a la IR canónica.

Deltas canónicos: 2591 sustituciones y 0 trazas registradas en las pasadas del álgebra (incluidas verificaciones); 0 deltas explícitos en las 13 cantidades finales. Detalle e índices sustituidos en `delta_contractions.json`.

Expresiones tensoriales abstractas

Las expresiones de esta sección son covariantes y no incorporan sustituciones del ansatz.

L

$$L = R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_3} + \frac{2}{\ell^2} + \beta_0 \ell^2 \left(-u_{d_0} g^{d_0 d_1} u_{d_1} R_{d_2 d_3 d_4 d_5} g^{d_2 d_4} g^{d_3 d_5} + 3 R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_4} g^{d_3 d_5} u_{d_4} u_{d_5} \right)$$

Significado: Densidad lagrangiana escalar normalizada de la teoría.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: validated_model

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

M_{ab}

$$M_{ab} = 2 R_{a d_0 b}{}^{d_0} + \beta_0 \ell^2 \left(-u_a u_b R_{d_0 d_1}{}^{d_0 d_1} - 3 R_{a d_0}{}^{d_0 d_1} u_b u_{d_1} - 3 u_a R_{b d_0}{}^{d_0 d_1} u_{d_1} - 2 R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} + 3 R_{a d_0 b}{}^{d_0 d_1} u_{d_0} u_{d_1} \right)$$

Significado: Derivada de L respecto de la métrica inversa, M_{ab} .

Índices libres y varianzas: $a_{\downarrow}, b_{\downarrow}$

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

P^{abcd}

$$P^{abcd} = \frac{1}{4} \left(-2 g^{ad} g^{bc} + 2 g^{ac} g^{bd} + \beta_0 \ell^2 \left(-3 g^{ad} u^b u^c - 3 u^a g^{bc} u^d - 2 g^{ac} g^{bd} u_{d_0} u^{d_0} \right. \right. \\ \left. \left. + 2 g^{ad} g^{bc} u_{d_0} u^{d_0} + 3 g^{ac} u^b u^d + 3 u^a g^{bd} u^c \right) \right)$$

Significado: Momento de curvatura $P^{abcd} = \partial^a L / \partial R_{abcd}$.

Índices libres y varianzas: $a^\uparrow, b^\uparrow, c^\uparrow, d^\uparrow$

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

J^a

$$J^a = -2 \beta_0 \ell^2 \left(u^a R_{d_0 d_1}{}^{d_0 d_1} + 3 R^a{}_{d_0}{}^{d_0 d_1} u_{d_1} \right)$$

Significado: Momento J^a conjugado a $nabla_a \phi$.

Índices libres y varianzas: $a^\uparrow$

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

F_ϕ

$$F_\phi = 0$$

Significado: Derivada explícita $F_\phi = \partial L / \partial \phi$.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: lagrangian

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

E_{ab}

$$\begin{aligned}
E_{ab} = & R_{a d_0 b}{}^{d_0} - \frac{g_{ab} R_{d_0 d_1}{}^{d_0 d_1}}{2} - \frac{g_{ab}}{\ell^2} + \frac{1}{4} \beta_0 \ell^2 \left(-12 R_{a d_0}{}^{d_0 d_1} u_b u_{d_1} - 12 u_a R_{b d_0}{}^{d_0 d_1} u_{d_1} \right. \\
& - 8 \nabla_{d_1}(u_{d_0}) \nabla_{d_2}(u^{d_1}) g_{ab} g^{d_0 d_2} - 6 \nabla_a(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{b d_1} - 6 \nabla_a(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{b d_1} \\
& - 6 \nabla_b(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{a d_1} - 6 \nabla_b(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{a d_1} - 6 g_{ab} R_{d_0}{}^{d_1 d_0 d_2} u_{d_1} u_{d_2} \\
& - 6 R_{a d_0 b d_1} u^{d_0} u^{d_1} - 4 \nabla_{d_2}(\nabla_{d_1}(u_{d_0})) g_{ab} g^{d_0 d_2} u^{d_1} - 4 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{ab} u_{d_0} g^{d_1 d_2} \\
& - 4 R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} - 4 u_a u_b R_{d_0 d_1}{}^{d_0 d_1} - 3 \nabla_a(\nabla_{d_0}(u^{d_0})) u_b - 3 \nabla_a(\nabla_{d_1}(u^{d_0})) g_{b d_0} u^{d_1} \\
& - 3 \nabla_b(\nabla_{d_0}(u^{d_0})) u_a - 3 \nabla_b(\nabla_{d_1}(u^{d_0})) g_{a d_0} u^{d_1} - 3 \nabla_{d_0}(\nabla_a(u^{d_0})) u_b - 3 \nabla_{d_0}(\nabla_b(u^{d_0})) u_a \\
& - 3 \nabla_{d_1}(\nabla_a(u^{d_0})) g_{b d_0} u^{d_1} - 3 \nabla_{d_1}(\nabla_b(u^{d_0})) g_{a d_0} u^{d_1} + 2 \nabla_a(\nabla_b(u^{d_0})) u_{d_0} \\
& + 2 \nabla_a(\nabla_{d_0}(u_b)) u^{d_0} + 2 \nabla_b(\nabla_a(u^{d_0})) u_{d_0} + 2 \nabla_b(\nabla_{d_0}(u_a)) u^{d_0} + 2 g_{ab} R_{d_0 d_1}{}^{d_0 d_1} u_{d_2} u^{d_2} \\
& + 3 R_{a d_0}{}^{d_0 d_1} u_b u^{d_1} + 3 u_a R_{b d_0}{}^{d_0 d_1} u^{d_1} + 4 \nabla_a(u^{d_0}) \nabla_{d_0}(u_b) + 4 \nabla_b(u^{d_0}) \nabla_{d_0}(u_a) \\
& + 6 \nabla_{d_0}(\nabla_{d_1}(u^{d_0})) g_{ab} u^{d_1} + 6 \nabla_{d_0}(u^{d_0}) \nabla_{d_1}(u^{d_1}) g_{ab} + 6 \nabla_{d_1}(\nabla_{d_0}(u^{d_0})) g_{ab} u^{d_1} \\
& + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{ab} + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_0} g_{b d_2} g^{d_1 d_3} \\
& + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_2} g_{b d_0} g^{d_1 d_3} + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{a d_0} u_b g^{d_1 d_2} \\
& \left. + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) u_a g_{b d_0} g^{d_1 d_2} + 12 R_{a}{}^{d_0}{}_{b}{}^{d_1} u_{d_0} u_{d_1} \right)
\end{aligned}$$

Significado: Ecuación de Euler-Lagrange métrica E_{ab} .

Índices libres y varianzas: $a_{\downarrow}, b_{\downarrow}$

Fuentes del pipeline: delta_lagrangian, curvature_derivative_metric_term

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$$\begin{aligned}
E_{ab}|_{\nabla\nabla P} = & \frac{1}{4} \beta_0 \ell^2 \left(-8 \nabla_{d_1}(u_{d_0}) \nabla_{d_2}(u^{d_1}) g_{ab} g^{d_0 d_2} - 6 \nabla_a(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{b d_1} \right. \\
& - 6 \nabla_a(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{b d_1} - 6 \nabla_b(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{a d_1} - 6 \nabla_b(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{a d_1} \\
& - 4 \nabla_{d_2}(\nabla_{d_1}(u_{d_0})) g_{ab} g^{d_0 d_2} u^{d_1} - 4 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{ab} u_{d_0} g^{d_1 d_2} - 3 \nabla_a(\nabla_{d_0}(u^{d_0})) u_b \\
& - 3 \nabla_a(\nabla_{d_1}(u^{d_0})) g_{b d_0} u^{d_1} - 3 \nabla_b(\nabla_{d_0}(u^{d_0})) u_a - 3 \nabla_b(\nabla_{d_1}(u^{d_0})) g_{a d_0} u^{d_1} \\
& - 3 \nabla_{d_0}(\nabla_a(u^{d_0})) u_b - 3 \nabla_{d_0}(\nabla_b(u^{d_0})) u_a - 3 \nabla_{d_1}(\nabla_a(u^{d_0})) g_{b d_0} u^{d_1} \\
& - 3 \nabla_{d_1}(\nabla_b(u^{d_0})) g_{a d_0} u^{d_1} + 2 \nabla_a(\nabla_b(u^{d_0})) u_{d_0} + 2 \nabla_a(\nabla_{d_0}(u_b)) u^{d_0} \\
& + 2 \nabla_b(\nabla_a(u^{d_0})) u_{d_0} + 2 \nabla_b(\nabla_{d_0}(u_a)) u^{d_0} + 4 \nabla_a(u^{d_0}) \nabla_{d_0}(u_b) + 4 \nabla_b(u^{d_0}) \nabla_{d_0}(u_a) \\
& + 6 \nabla_{d_0}(\nabla_{d_1}(u^{d_0})) g_{ab} u^{d_1} + 6 \nabla_{d_0}(u^{d_0}) \nabla_{d_1}(u^{d_1}) g_{ab} + 6 \nabla_{d_1}(\nabla_{d_0}(u^{d_0})) g_{ab} u^{d_1} \\
& + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{ab} + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_0} g_{b d_2} g^{d_1 d_3} \\
& + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_2} g_{b d_0} g^{d_1 d_3} + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{a d_0} u_b g^{d_1 d_2} \\
& \left. + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) u_a g_{b d_0} g^{d_1 d_2} \right)
\end{aligned}$$

La expresión anterior identifica la contribución $-2\nabla^c\nabla^d P_{acdb}$ dentro de E_{ab} .

E_ϕ

$$E_\phi = 2\beta_0\ell^2\left(\nabla_{d_0}\left(u^{d_0}\right)R_{d_1d_2}{}^{d_1d_2} + \nabla_{d_2}\left(R_{d_0d_1}{}^{d_0d_1}\right)u^{d_2} - 3\nabla_{d_2}(u_{d_1})R_{d_0}{}^{d_2d_0d_1} + 3\nabla_{d_1}\left(-R_{d_0}{}^{d_1d_0d_2}\right)u_{d_2}\right)$$

Significado: Ecuación de Euler-Lagrange escalar E_ϕ .

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: `scalar_derivative`, `scalar_gradient_momentum`

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$\mathcal{R}$

$$\mathcal{R} = g^{ac}g^{bd}R_{abcd}$$

Significado: Escalar de Ricci construido con la convención geométrica activa.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: `validated_model`

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$R_{ab}R^{ab}$

$$R_{ab}R^{ab} = g^{ac}R_{abcd}g^{be}g^{df}g^{gh}R_{ghef}$$

Significado: Invariante cuadrático de Ricci $R_{ab}R^{ab}$.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: `validated_model`, `riemann_tensor`

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

R_{abcd}

$$R_{abcd} = R_{abcd}$$

Significado: Tensor de Riemann covariante tratado como entrada geométrica.

Índices libres y varianzas: $a_\downarrow, b_\downarrow, c_\downarrow, d_\downarrow$

Fuentes del pipeline: `validated_model`

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$R_{abcd}R^{abcd}$

$$R_{abcd}R^{abcd} = g^{ae} g^{bf} g^{cg} g^{dh} R_{a b c d} R_{e f g h}$$

Significado: Invariante de Kretschmann $R_{abcd} R^{abcd}$.

Índices libres y varianzas: $\emptyset$ (escalar)

Fuentes del pipeline: `validated_model`, `riemann_tensor`

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$\nabla_e P^{abcd}$

$$\begin{aligned} \nabla_e P^{abcd} = \frac{1}{4} \beta_0 \ell^2 \big(& -3 \nabla_e(u^a) g^{bc} u^d - 3 \nabla_e(u^b) g^{ad} u^c - 3 \nabla_e(u^c) g^{ad} u^b - 3 \nabla_e(u^d) u^a g^{bc} \\ & - 2 \nabla_{d_0}(u_e) g^{ac} g^{bd} u^{d_0} - 2 \nabla_e(u^{d_0}) g^{ac} g^{bd} u_{d_0} + 2 \nabla_{d_0}(u_e) g^{ad} g^{bc} u^{d_0} \\ & + 2 \nabla_e(u^{d_0}) g^{ad} g^{bc} u_{d_0} + 3 \nabla_e(u^a) g^{bd} u^c + 3 \nabla_e(u^b) g^{ac} u^d + 3 \nabla_e(u^c) u^a g^{bd} \\ & + 3 \nabla_e(u^d) g^{ac} u^b \big) \end{aligned}$$

Significado: Primera derivada covariante del momento de curvatura.

Índices libres y varianzas: $a^\uparrow, b^\uparrow, c^\uparrow, d^\uparrow, e_\downarrow$

Fuentes del pipeline: `curvature_momentum`

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

$\nabla_f \nabla_e P^{abcd}$

$$\begin{aligned} \nabla_f \nabla_e P^{abcd} = \frac{1}{4} \beta_0 \ell^2 \big(& -3 \nabla_e(u^a) \nabla_f(u^d) g^{bc} - 3 \nabla_e(u^b) \nabla_f(u^c) g^{ad} - 3 \nabla_e(u^c) \nabla_f(u^b) g^{ad} \\ & - 3 \nabla_e(u^d) \nabla_f(u^a) g^{bc} - 3 \nabla_f(\nabla_e(u^a)) g^{bc} u^d - 3 \nabla_f(\nabla_e(u^b)) g^{ad} u^c \\ & - 3 \nabla_f(\nabla_e(u^c)) g^{ad} u^b - 3 \nabla_f(\nabla_e(u^d)) u^a g^{bc} - 2 \nabla_{d_0}(u_e) \nabla_f(u^{d_0}) g^{ac} g^{bd} \\ & - 2 \nabla_{d_0}(u_f) \nabla_e(u^{d_0}) g^{ac} g^{bd} - 2 \nabla_f(\nabla_{d_0}(u_e)) g^{ac} g^{bd} u^{d_0} - 2 \nabla_f(\nabla_e(u^{d_0})) g^{ac} g^{bd} u_{d_0} \\ & + 2 \nabla_{d_0}(u_e) \nabla_f(u^{d_0}) g^{ad} g^{bc} + 2 \nabla_{d_0}(u_f) \nabla_e(u^{d_0}) g^{ad} g^{bc} + 2 \nabla_f(\nabla_{d_0}(u_e)) g^{ad} g^{bc} u^{d_0} \\ & + 2 \nabla_f(\nabla_e(u^{d_0})) g^{ad} g^{bc} u_{d_0} + 3 \nabla_e(u^a) \nabla_f(u^c) g^{bd} + 3 \nabla_e(u^b) \nabla_f(u^d) g^{ac} \\ & + 3 \nabla_e(u^c) \nabla_f(u^a) g^{bd} + 3 \nabla_e(u^d) \nabla_f(u^b) g^{ac} + 3 \nabla_f(\nabla_e(u^a)) g^{bd} u^c \\ & + 3 \nabla_f(\nabla_e(u^b)) g^{ac} u^d + 3 \nabla_f(\nabla_e(u^c)) u^a g^{bd} + 3 \nabla_f(\nabla_e(u^d)) g^{ac} u^b \big) \end{aligned}$$

Significado: Segunda derivada covariante del momento de curvatura.

Índices libres y varianzas: $a^\uparrow, b^\uparrow, c^\uparrow, d^\uparrow, e_\downarrow, f_\downarrow$

Fuentes del pipeline: `nabla_P`

Wolfram/xAct: validación xAct no solicitada. La corrida no solicitó validación Wolfram/xAct.

Descomposición compacta adicional

Esta vista se agrega después de los resultados anteriores y reutiliza exactamente la IR canónica ya calculada. No interviene en la validación ni en los fingerprints.

Descomposición de E_{ab} .

$$E_{ab} = M_{ab} - \mathcal{R}_{(ab)}[P] - \frac{1}{2}g_{ab}L - 2\nabla_m \nabla_n P^{mn}{}_{ab}$$

Reconstrucción IR: verified. Los cuatro bloques reconstruyen exactamente E_{ab} en la IR canónica.

forma compacta:
$$E_{ab} = R_{a d_0 b}{}^{d_0} - \frac{g_{ab} R_{d_0 d_1}{}^{d_0 d_1}}{2} - \frac{g_{ab}}{\ell^2} + \frac{1}{4} \beta_0 \ell^2 \left(-12 R_{a d_0}{}^{d_0 d_1} u_b u_{d_1} \right. \\ - 12 u_a R_{b d_0}{}^{d_0 d_1} u_{d_1} - 8 \nabla_{d_1}(u_{d_0}) \nabla_{d_2}(u^{d_1}) g_{ab} g^{d_0 d_2} \\ - 6 \nabla_a(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{b d_1} - 6 \nabla_a(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{b d_1} \\ - 6 \nabla_b(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{a d_1} - 6 \nabla_b(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{a d_1} \\ - 6 g_{ab} R_{d_0}{}^{d_1 d_0 d_2} u_{d_1} u_{d_2} - 6 R_{a d_0 b d_1} u^{d_0} u^{d_1} - 4 \nabla_{d_2}(\nabla_{d_1}(u_{d_0})) g_{ab} g^{d_0 d_2} u^{d_1} \\ - 4 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{ab} u_{d_0} g^{d_1 d_2} - 4 R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} - 4 u_a u_b R_{d_0 d_1}{}^{d_0 d_1} \\ - 3 \nabla_a(\nabla_{d_0}(u^{d_0})) u_b - 3 \nabla_a(\nabla_{d_1}(u^{d_0})) g_{b d_0} u^{d_1} - 3 \nabla_b(\nabla_{d_0}(u^{d_0})) u_a \\ - 3 \nabla_b(\nabla_{d_1}(u^{d_0})) g_{a d_0} u^{d_1} - 3 \nabla_{d_0}(\nabla_a(u^{d_0})) u_b - 3 \nabla_{d_0}(\nabla_b(u^{d_0})) u_a \\ - 3 \nabla_{d_1}(\nabla_a(u^{d_0})) g_{b d_0} u^{d_1} - 3 \nabla_{d_1}(\nabla_b(u^{d_0})) g_{a d_0} u^{d_1} \\ + 2 \nabla_a(\nabla_b(u^{d_0})) u_{d_0} + 2 \nabla_a(\nabla_{d_0}(u_b)) u^{d_0} + 2 \nabla_b(\nabla_a(u^{d_0})) u_{d_0} \\ + 2 \nabla_b(\nabla_{d_0}(u_a)) u^{d_0} + 2 g_{ab} R_{d_0 d_1}{}^{d_0 d_1} u_{d_2} u^{d_2} + 3 R_{a d_0}{}^{d_0 d_1} u_b u^{d_1} \\ + 3 u_a R_{b d_0}{}^{d_0 d_1} u^{d_1} + 4 \nabla_a(u^{d_0}) \nabla_{d_0}(u_b) + 4 \nabla_b(u^{d_0}) \nabla_{d_0}(u_a) \\ + 6 \nabla_{d_0}(\nabla_{d_1}(u^{d_0})) g_{ab} u^{d_1} + 6 \nabla_{d_0}(u^{d_0}) \nabla_{d_1}(u^{d_1}) g_{ab} \\ + 6 \nabla_{d_1}(\nabla_{d_0}(u^{d_0})) g_{ab} u^{d_1} + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{ab} \\ + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_0} g_{b d_2} g^{d_1 d_3} \\ + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_2} g_{b d_0} g^{d_1 d_3} + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{a d_0} u_b g^{d_1 d_2} \\ \left. + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) u_a g_{b d_0} g^{d_1 d_2} + 12 R_{a d_0 b}{}^{d_0 d_1} u_{d_0} u_{d_1} \right)$$

Bloque M_{ab} (fuentes: `metric_momentum`).

forma compacta:
$$M_{ab} = 2 R_{a d_0 b}{}^{d_0} + \beta_0 \ell^2 \left(-u_a u_b R_{d_0 d_1}{}^{d_0 d_1} - 3 R_{a d_0}{}^{d_0 d_1} u_b u_{d_1} - 3 u_a R_{b d_0}{}^{d_0 d_1} u_{d_1} \right. \\ \left. - 2 R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} + 3 R_{a d_0 b}{}^{d_0 d_1} u_{d_0} u_{d_1} \right)$$

Bloque $-\mathcal{R}_{(ab)}[P]$ (fuentes: `curvature_momentum`, `riemann_tensor`).

forma compacta:
$$-\mathcal{R}_{(ab)}[P] = \frac{1}{2} \left(-g_{a d_1} \left(-\frac{\ell^2 g^{d_1 d_2} g^{d_3 d_4} u_{d_0} u^{d_0} \beta_0}{2} - \frac{g^{d_1 d_4} g^{d_3 d_2}}{2} \right. \right. \\ + \frac{\ell^2 g^{d_1 d_4} g^{d_3 d_2} u_{d_0} u^{d_0} \beta_0}{2} + \frac{g^{d_1 d_2} g^{d_3 d_4}}{2} - \frac{3 \ell^2 g^{d_1 d_4} u^{d_3} u^{d_2} \beta_0}{4} \\ - \frac{3 \ell^2 u^{d_1} g^{d_3 d_2} u^{d_4} \beta_0}{4} + \frac{3 \ell^2 g^{d_1 d_2} u^{d_3} u^{d_4} \beta_0}{4} \\ + \frac{3 \ell^2 u^{d_1} g^{d_3 d_4} u^{d_2} \beta_0}{4} \Big) R_{b d_3 d_2 d_4} - g_{b d_1} \left(-\frac{\ell^2 g^{d_1 d_2} g^{d_3 d_4} u_{d_0} u^{d_0} \beta_0}{2} \right. \\ - \frac{g^{d_1 d_4} g^{d_3 d_2}}{2} + \frac{\ell^2 g^{d_1 d_4} g^{d_3 d_2} u_{d_0} u^{d_0} \beta_0}{2} + \frac{g^{d_1 d_2} g^{d_3 d_4}}{2} \\ - \frac{3 \ell^2 g^{d_1 d_4} u^{d_3} u^{d_2} \beta_0}{4} - \frac{3 \ell^2 u^{d_1} g^{d_3 d_2} u^{d_4} \beta_0}{4} + \frac{3 \ell^2 g^{d_1 d_2} u^{d_3} u^{d_4} \beta_0}{4} \\ \left. \left. + \frac{3 \ell^2 u^{d_1} g^{d_3 d_4} u^{d_2} \beta_0}{4} \right) R_{a d_3 d_2 d_4} \right)$$

Bloque $-\frac{1}{2}g_{ab}L$ (fuentes: `lagrangian`).

forma compacta:
$$-\frac{1}{2}g_{ab}L = -\frac{1}{2}g_{ab} \left(-\ell^2 u_{d_0} g^{d_0 d_1} u_{d_1} R_{d_2 d_3 d_4 d_5} g^{d_2 d_4} g^{d_3 d_5} \beta_0 + \frac{2}{\ell^2} \right. \\ \left. + 3 \ell^2 R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_4} g^{d_3 d_5} u_{d_4} u_{d_5} \beta_0 + R_{d_0 d_1 d_2 d_3} g^{d_0 d_2} g^{d_1 d_3} \right)$$

Bloque $E_{ab}^{(\nabla\nabla P)}$ (fuentes: `curvature_momentum`, `nabla_nabla_P`).

forma compacta:
$$E_{ab}^{(\nabla\nabla P)} = \frac{1}{4} \beta_0 \ell^2 \left(-8 \nabla_{d_1}(u_{d_0}) \nabla_{d_2}(u^{d_1}) g_{ab} g^{d_0 d_2} - 6 \nabla_a(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{b d_1} \right. \\ - 6 \nabla_a(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{b d_1} - 6 \nabla_b(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{a d_1} \\ - 6 \nabla_b(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{a d_1} - 4 \nabla_{d_2}(\nabla_{d_1}(u_{d_0})) g_{ab} g^{d_0 d_2} u^{d_1} \\ - 4 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{ab} u_{d_0} g^{d_1 d_2} - 3 \nabla_a(\nabla_{d_0}(u^{d_0})) u_b \\ - 3 \nabla_a(\nabla_{d_1}(u^{d_0})) g_{b d_0} u^{d_1} - 3 \nabla_b(\nabla_{d_0}(u^{d_0})) u_a \\ - 3 \nabla_b(\nabla_{d_1}(u^{d_0})) g_{a d_0} u^{d_1} - 3 \nabla_{d_0}(\nabla_a(u^{d_0})) u_b \\ - 3 \nabla_{d_0}(\nabla_b(u^{d_0})) u_a - 3 \nabla_{d_1}(\nabla_a(u^{d_0})) g_{b d_0} u^{d_1} \\ - 3 \nabla_{d_1}(\nabla_b(u^{d_0})) g_{a d_0} u^{d_1} + 2 \nabla_a(\nabla_b(u^{d_0})) u_{d_0} + 2 \nabla_a(\nabla_{d_0}(u_b)) u^{d_0} \\ + 2 \nabla_b(\nabla_a(u^{d_0})) u_{d_0} + 2 \nabla_b(\nabla_{d_0}(u_a)) u^{d_0} + 4 \nabla_a(u^{d_0}) \nabla_{d_0}(u_b) \\ + 4 \nabla_b(u^{d_0}) \nabla_{d_0}(u_a) + 6 \nabla_{d_0}(\nabla_{d_1}(u^{d_0})) g_{ab} u^{d_1} \\ + 6 \nabla_{d_0}(u^{d_0}) \nabla_{d_1}(u^{d_1}) g_{ab} + 6 \nabla_{d_1}(\nabla_{d_0}(u^{d_0})) g_{ab} u^{d_1} \\ + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{ab} + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_0} g_{b d_2} g^{d_1 d_3} \\ + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_2} g_{b d_0} g^{d_1 d_3} \\ \left. + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{a d_0} u_b g^{d_1 d_2} + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) u_a g_{b d_0} g^{d_1 d_2} \right)$$

Descomposición de E_ϕ .

$$E_\phi = F_\phi - \nabla_a J^a$$

Reconstrucción IR: verified. Los dos bloques reconstruyen exactamente E_ϕ en la IR canónica.

forma compacta:
$$E_\phi = 2\beta_0 \ell^2 \left(\nabla_{d_0} \left(u^{d_0} \right) R_{d_1 d_2}{}^{d_1 d_2} + \nabla_{d_2} \left(R_{d_0 d_1}{}^{d_0 d_1} \right) u^{d_2} - 3 \nabla_{d_2} (u_{d_1}) R_{d_0}{}^{d_2 d_0 d_1} \right. \\ \left. + 3 \nabla_{d_1} \left(-R_{d_0}{}^{d_1 d_0 d_2} \right) u_{d_2} \right)$$

Bloque F_ϕ (fuentes: `scalar_derivative`).

forma compacta:
$$F_\phi = 0$$

Bloque $-\nabla_a J^a$ (fuentes: `scalar_gradient_momentum`, `scalar_euler`).

forma compacta:
$$-\nabla_a J^a = 2\beta_0 \ell^2 \left(\nabla_{d_0} \left(u^{d_0} \right) R_{d_1 d_2}{}^{d_1 d_2} + \nabla_{d_2} \left(R_{d_0 d_1}{}^{d_0 d_1} \right) u^{d_2} \right. \\ \left. - 3 \nabla_{d_2} (u_{d_1}) R_{d_0}{}^{d_2 d_0 d_1} + 3 \nabla_{d_1} \left(-R_{d_0}{}^{d_1 d_0 d_2} \right) u_{d_2} \right)$$

Descomposición de P^{abcd} .

$$P^{abcd} = \partial L / \partial R_{abcd}$$

Reconstrucción IR: verified. La forma compacta y la expandida referencian el mismo objeto P^{abcd} .

forma compacta:
$$P^{abcd} = \frac{1}{4} \left(-2 g^{ad} g^{bc} + 2 g^{ac} g^{bd} + \beta_0 \ell^2 \left(-3 g^{ad} u^b u^c - 3 u^a g^{bc} u^d \right. \right. \\ \left. \left. - 2 g^{ac} g^{bd} u_{d_0} u^{d_0} + 2 g^{ad} g^{bc} u_{d_0} u^{d_0} + 3 g^{ac} u^b u^d + 3 u^a g^{bd} u^c \right) \right)$$

Expresiones proyectadas mediante el ansatz

Ansatz utilizado: `draft4_circular`.

$$(x^\mu) = (\tau, r, \varphi), \quad ds^2 \\ = -f(r) d\tau^2 + \frac{1}{f(r)} dr^2 + r^2 d\varphi^2$$

$$\phi = \Phi(r, \varphi) \quad (\text{genérico, sin especialización})$$

Las componentes completas se conservan de forma dispersa en `results.json`; el documento muestra como máximo doce componentes no nulas por tensor.

L

Ansatz: `draft4_circular`; **estado:** completada.

$$L|_{\text{ansatz}} = \frac{1}{2r^3} \left(-4 f'(r) r^2 - 2 f''(r) r^3 + \frac{4r^3}{\ell^2} + \beta_0 \ell^2 \left(-2 f'(r) \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \right)^2 + 2 f''(r) r \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) \right)^2 \right. \right. \\ \left. \left. + f'(r) f(r) r^2 \left(\frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \right)^2 - f''(r) f(r) r^3 \left(\frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \right)^2 \right) \right)$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz '`draft4_circular`'.

M_{ab}

Ansatz: draft4_circular; **estado:** completada.

$$[M_{ab}]_{00} = \frac{1}{2r^3} \left(2r^2 (r f''(r) + f'(r)) + \beta_0 \ell^2 \left(-2 \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) (r f''(r) + f'(r)) \right. \right. \\ \left. \left. + 3 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 f'(r) + 3r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f''(r) f(r) \right) \right) f(r)$$

$$[M_{ab}]_{11} = \frac{1}{2r^3 f(r)} \left(-2 f'(r) r^2 - 2 f''(r) r^3 \right. \\ \left. + \beta_0 \ell^2 \left(-f'(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + 2r f''(r) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 - 2 f(r) f''(r) r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 \right) \right)$$

$$[M_{ab}]_{12} = -\frac{1}{2r} \beta_0 \ell^2 (2 f'(r) + r f''(r)) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$[M_{ab}]_{21} = -\frac{1}{2r} \beta_0 \ell^2 (2 f'(r) + r f''(r)) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$[M_{ab}]_{22} = -2r f'(r) + \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 f''(r) - \frac{2 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 f'(r)}{r} + \frac{\beta_0 r \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) f(r)}{2}$$

Proyección completa almacenada: 5 componentes no nulas de 9. Proyectada respetando el ansatz 'draft4_circular'.

P^{abcd}

Ansatz: draft4_circular; **estado:** completada.

$$[P^{abcd}]^{0101} = \frac{\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2}{2r^2} - \frac{\beta_0 f(r) \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2}{4} - \frac{1}{2}$$

$$[P^{abcd}]^{0102} = -\frac{1}{4r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$[P^{abcd}]^{0201} = -\frac{1}{4r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$[P^{abcd}]^{0202} = \frac{1}{4r^4 f(r)} \left(-2r^2 + \beta_0 \ell^2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + 2r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P^{abcd}\right]^{0110} = -\frac{\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2}{2 r^2} + \frac{\beta_0 f(r) \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2}{4} + \frac{1}{2}$$

$$\left[P^{abcd}\right]^{0210} = \frac{1}{4 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$\left[P^{abcd}\right]^{0120} = \frac{1}{4 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$\left[P^{abcd}\right]^{0220} = \frac{1}{4 r^4 f(r)} \left(2 r^2 - \beta_0 \ell^2 \left(-\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + 2 r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f(r) \right) \right)$$

$$\left[P^{abcd}\right]^{1001} = -\frac{\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2}{2 r^2} + \frac{\beta_0 f(r) \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2}{4} + \frac{1}{2}$$

$$\left[P^{abcd}\right]^{1002} = \frac{1}{4 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)$$

$$\left[P^{abcd}\right]^{1010} = \frac{\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2}{2 r^2} - \frac{\beta_0 f(r) \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2}{4} - \frac{1}{2}$$

$$\left[P^{abcd}\right]^{1212} = \frac{1}{4 r^4} f(r) \left(2 r^2 + \beta_0 \ell^2 \left(\frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)^2 + f(r) r^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 \right) \right)$$

Representación compacta: se muestran 12 de 20 componentes no nulas. Proyección completa almacenada: 20 componentes no nulas de 81. Proyectada respetando el ansatz 'draft4_circular'.

J^a

Ansatz: draft4_circular; **estado:** completada.

$$[J^a]^1 = \frac{1}{r} \beta_0 \ell^2 \left(-r f''(r) + f'(r) \right) \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f(r)$$

$$[J^a]^2 = \frac{2 \beta_0 \ell^2 \left(-f'(r) + r f''(r) \right) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi)}{r^3}$$

Proyección completa almacenada: 2 componentes no nulas de 3. Proyectada respetando el ansatz 'draft4_circular'.

F_ϕ

Ansatz: draft4_circular; **estado:** completada.

$$F_\phi|_{\text{ansatz}} = 0$$

Proyección completa almacenada: 0 componentes no nulas de 1. La expresión abstracta es nula; todas sus componentes son cero.

E_{ab}

Ansatz: draft4_circular; **estado:** no disponible por limitación del backend.

La forma abstracta se conserva sin sustitución:

$$\begin{aligned} E_{ab} = & R_{a d_0 b}{}^{d_0} - \frac{g_{a b} R_{d_0 d_1}{}^{d_0 d_1}}{2} - \frac{g_{a b}}{\ell^2} + \frac{1}{4} \beta_0 \ell^2 \left(-12 R_{a d_0}{}^{d_0 d_1} u_b u_{d_1} - 12 u_a R_{b d_0}{}^{d_0 d_1} u_{d_1} \right. \\ & - 8 \nabla_{d_1}(u_{d_0}) \nabla_{d_2}(u^{d_1}) g_{a b} g^{d_0 d_2} - 6 \nabla_a(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{b d_1} - 6 \nabla_a(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{b d_1} \\ & - 6 \nabla_b(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{a d_1} - 6 \nabla_b(u^{d_1}) \nabla_{d_0}(u^{d_0}) g_{a d_1} - 6 g_{a b} R_{d_0}{}^{d_1 d_0 d_2} u_{d_1} u_{d_2} \\ & - 6 R_{a d_0 b d_1} u^{d_0} u^{d_1} - 4 \nabla_{d_2}(\nabla_{d_1}(u_{d_0})) g_{a b} g^{d_0 d_2} u^{d_1} - 4 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{a b} u_{d_0} g^{d_1 d_2} \\ & - 4 R_{a d_0 b}{}^{d_0} u_{d_1} u^{d_1} - 4 u_a u_b R_{d_0 d_1}{}^{d_0 d_1} - 3 \nabla_a(\nabla_{d_0}(u^{d_0})) u_b - 3 \nabla_a(\nabla_{d_1}(u^{d_0})) g_{b d_0} u^{d_1} \\ & - 3 \nabla_b(\nabla_{d_0}(u^{d_0})) u_a - 3 \nabla_b(\nabla_{d_1}(u^{d_0})) g_{a d_0} u^{d_1} - 3 \nabla_{d_0}(\nabla_a(u^{d_0})) u_b - 3 \nabla_{d_0}(\nabla_b(u^{d_0})) u_a \\ & - 3 \nabla_{d_1}(\nabla_a(u^{d_0})) g_{b d_0} u^{d_1} - 3 \nabla_{d_1}(\nabla_b(u^{d_0})) g_{a d_0} u^{d_1} + 2 \nabla_a(\nabla_b(u^{d_0})) u_{d_0} \\ & + 2 \nabla_a(\nabla_{d_0}(u_b)) u^{d_0} + 2 \nabla_b(\nabla_a(u^{d_0})) u_{d_0} + 2 \nabla_b(\nabla_{d_0}(u_a)) u^{d_0} + 2 g_{a b} R_{d_0 d_1}{}^{d_0 d_1} u_{d_2} u^{d_2} \\ & + 3 R_{a d_0}{}^{d_0 d_1} u_b u^{d_1} + 3 u_a R_{b d_0}{}^{d_0 d_1} u^{d_1} + 4 \nabla_a(u^{d_0}) \nabla_{d_0}(u_b) + 4 \nabla_b(u^{d_0}) \nabla_{d_0}(u_a) \\ & + 6 \nabla_{d_0}(\nabla_{d_1}(u^{d_0})) g_{a b} u^{d_1} + 6 \nabla_{d_0}(u^{d_0}) \nabla_{d_1}(u^{d_1}) g_{a b} + 6 \nabla_{d_1}(\nabla_{d_0}(u^{d_0})) g_{a b} u^{d_1} \\ & + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_0}(u^{d_1}) g_{a b} + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_0} g_{b d_2} g^{d_1 d_3} \\ & + 6 \nabla_{d_1}(u^{d_0}) \nabla_{d_3}(u^{d_2}) g_{a d_2} g_{b d_0} g^{d_1 d_3} + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) g_{a d_0} u_b g^{d_1 d_2} \\ & \left. + 6 \nabla_{d_2}(\nabla_{d_1}(u^{d_0})) u_a g_{b d_0} g^{d_1 d_2} + 12 R_{a d_0 b}{}^{d_0 d_1} u_{d_0} u_{d_1} \right) \end{aligned}$$

Motivo: La expresión se conserva simbólica: su proyección con el campo genérico $\text{Phi}(r, \text{varphi})$ contiene 58 nodos de derivada covariante, por encima del límite seguro 24 del backend. Especialice el perfil escalar o eleve el límite del backend de componentes.

E_ϕ

Ansatz: draft4_circular; **estado:** completada.

$$\begin{aligned} E_\phi|_{\text{ansatz}} = & \frac{1}{r^3} \beta_0 \ell^2 \left(2 \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi) f'(r) - r^2 f'(r)^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) - 2 r \frac{\partial^2 \Phi}{\partial \varphi^2}(r, \varphi) f''(r) \right. \\ & + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f'(r) f''(r) + r^3 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) f^{(3)}(r) f(r) + r^3 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f''(r) f(r) \\ & \left. - r^2 \frac{\partial^2 \Phi}{\partial r^2}(r, \varphi) f'(r) f(r) \right) \end{aligned}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'.

$\mathcal{R}$

Ansatz: draft4_circular; **estado:** completada.

$$\mathcal{R}|_{\text{ansatz}} = - \left(\frac{2 f'(r)}{r} + f''(r) \right)$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$R_{ab}R^{ab}$

Ansatz: draft4_circular; **estado:** completada.

$$R_{ab}R^{ab}|_{\text{ansatz}} = \frac{f''(r)^2}{2} + \frac{3 f'(r)^2}{2 r^2} + \frac{f'(r) f''(r)}{r}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

R_{abcd}

Ansatz: draft4_circular; **estado:** completada.

$$[R_{abcd}]_{0101} = \frac{f''(r)}{2}$$

$$[R_{abcd}]_{0110} = -\frac{f''(r)}{2}$$

$$[R_{abcd}]_{0202} = \frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{0220} = -\frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{1001} = -\frac{f''(r)}{2}$$

$$[R_{abcd}]_{1010} = \frac{f''(r)}{2}$$

$$[R_{abcd}]_{1212} = -\frac{r f'(r)}{2 f(r)}$$

$$[R_{abcd}]_{1221} = \frac{r f'(r)}{2 f(r)}$$

$$[R_{abcd}]_{2002} = -\frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{2020} = \frac{r f'(r) f(r)}{2}$$

$$[R_{abcd}]_{2112} = \frac{r f'(r)}{2 f(r)}$$

$$[R_{abcd}]_{2121} = -\frac{r f'(r)}{2 f(r)}$$

Proyección completa almacenada: 12 componentes no nulas de 81. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$$R_{abcd}R^{abcd}$$

Ansatz: draft4_circular; **estado:** completada.

$$R_{abcd}R^{abcd}|_{\text{ansatz}} = f''(r)^2 + \frac{2 f'(r)^2}{r^2}$$

Proyección completa almacenada: 1 componentes no nulas de 1. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$$\nabla_e P^{abcd}$$

Ansatz: draft4_circular; **estado:** completada.

$$\left[\nabla_e P^{abcd} \right]_0^{0112} = -\frac{1}{8 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) f(r)$$

$$\left[\nabla_e P^{abcd} \right]_0^{0212} = \frac{3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) f(r)}{8 r^2}$$

$$\left[\nabla_e P^{abcd} \right]_0^{0121} = \frac{1}{8 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) f(r)$$

$$\left[\nabla_e P^{abcd} \right]_0^{0221} = -\frac{3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) f(r)}{8 r^2}$$

$$\left[\nabla_e P^{abcd} \right]_0^{1201} = -\frac{1}{8 r^2} 3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) f(r)$$

$$\left[\nabla_e P^{abcd} \right]_0^{1202} = \frac{3 \beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) f(r)}{8 r^2}$$

$$\left[\nabla_e P^{abcd}\right]_0^{1012} = \frac{1}{8r^2} 3\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) f(r)$$

$$\left[\nabla_e P^{abcd}\right]_0^{1210} = \frac{1}{8r^2} 3\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) f(r)$$

$$\left[\nabla_e P^{abcd}\right]_0^{1021} = -\frac{1}{8r^2} 3\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) f(r)$$

$$\left[\nabla_e P^{abcd}\right]_0^{1220} = -\frac{3\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) f(r)}{8r^2}$$

$$\left[\nabla_e P^{abcd}\right]_0^{2101} = \frac{1}{8r^2} 3\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi) \frac{\partial^1 \Phi}{\partial \varphi^1}(r, \varphi) f'(r) f(r)$$

$$\left[\nabla_e P^{abcd}\right]_0^{2102} = -\frac{3\beta_0 \ell^2 \frac{\partial^1 \Phi}{\partial r^1}(r, \varphi)^2 f'(r) f(r)}{8r^2}$$

Representación compacta: se muestran 12 de 56 componentes no nulas. Proyección completa almacenada: 56 componentes no nulas de 243. Proyectada respetando el ansatz 'draft4_circular'. La corrida no solicitó validación Wolfram/xAct.

$$\nabla_f \nabla_e P^{abcd}$$

Ansatz: draft4_circular; **estado:** no disponible por limitación del backend.

La forma abstracta se conserva sin sustitución:

$$\begin{aligned} \nabla_f \nabla_e P^{abcd} = & \frac{1}{4} \beta_0 \ell^2 \left(-3 \nabla_e(u^a) \nabla_f(u^d) g^{bc} - 3 \nabla_e(u^b) \nabla_f(u^c) g^{ad} - 3 \nabla_e(u^c) \nabla_f(u^b) g^{ad} \right. \\ & - 3 \nabla_e(u^d) \nabla_f(u^a) g^{bc} - 3 \nabla_f(\nabla_e(u^a)) g^{bc} u^d - 3 \nabla_f(\nabla_e(u^b)) g^{ad} u^c \\ & - 3 \nabla_f(\nabla_e(u^c)) g^{ad} u^b - 3 \nabla_f(\nabla_e(u^d)) u^a g^{bc} - 2 \nabla_{d_0}(u_e) \nabla_f(u^{d_0}) g^{ac} g^{bd} \\ & - 2 \nabla_{d_0}(u_f) \nabla_e(u^{d_0}) g^{ac} g^{bd} - 2 \nabla_f(\nabla_{d_0}(u_e)) g^{ac} g^{bd} u^{d_0} - 2 \nabla_f(\nabla_e(u^{d_0})) g^{ac} g^{bd} u_{d_0} \\ & + 2 \nabla_{d_0}(u_e) \nabla_f(u^{d_0}) g^{ad} g^{bc} + 2 \nabla_{d_0}(u_f) \nabla_e(u^{d_0}) g^{ad} g^{bc} + 2 \nabla_f(\nabla_{d_0}(u_e)) g^{ad} g^{bc} u^{d_0} \\ & + 2 \nabla_f(\nabla_e(u^{d_0})) g^{ad} g^{bc} u_{d_0} + 3 \nabla_e(u^a) \nabla_f(u^c) g^{bd} + 3 \nabla_e(u^b) \nabla_f(u^d) g^{ac} \\ & + 3 \nabla_e(u^c) \nabla_f(u^a) g^{bd} + 3 \nabla_e(u^d) \nabla_f(u^b) g^{ac} + 3 \nabla_f(\nabla_e(u^a)) g^{bd} u^c \\ & \left. + 3 \nabla_f(\nabla_e(u^b)) g^{ac} u^d + 3 \nabla_f(\nabla_e(u^c)) u^a g^{bd} + 3 \nabla_f(\nabla_e(u^d)) g^{ac} u^b \right) \end{aligned}$$

Motivo: La expresión se conserva simbólica: su proyección con el campo genérico Phi(r, varphi) contiene 48 nodos de derivada covariante, por encima del límite seguro 24 del backend. Especialice el perfil escalar o eleve el límite del backend de componentes. La corrida no solicitó validación Wolfram/xAct.

Extras: perfiles escalares de una variable, con la misma métrica del ansatz genérico. Se sustituyen únicamente las componentes de L y P^{abcd} ya calculadas; sin validación xAct independiente de estos perfiles. Las componentes completas de estos extras constan en `presentation.json`.

Extra: $\phi = \Phi(r)$

$$L = \frac{1}{2\ell^2 r} \left(4r + \ell^2 \left(f'(r) \left(\beta_0 f(r) \ell^2 \Phi'(r)^2 - 4 \right) - r f''(r) \left(\beta_0 f(r) \ell^2 \Phi'(r)^2 + 2 \right) \right) \right)$$

Componentes independientes no nulas; $P^{abcd} = -P^{bacd} = -P^{abdc} = P^{cdab}$.

$$\left[P^{abcd} \right]^{0101} = -\frac{\beta_0 f(r) \ell^2 \Phi'(r)^2}{4} - \frac{1}{2}$$

$$\left[P^{abcd} \right]^{0202} = \frac{\left(\beta_0 \ell^2 \Phi'(r)^2 f(r) - 1 \right)}{2r^2 f(r)}$$

$$\left[P^{abcd} \right]^{1212} = \frac{\left(\beta_0 \ell^2 \Phi'(r)^2 f(r) + 2 \right) f(r)}{4r^2}$$

Extra: $\phi = \Phi(\varphi)$

$$L = \frac{1}{r^3} \left(-f''(r) r^3 - 2f'(r) r^2 + \frac{2r^3}{\ell^2} + \beta_0 \ell^2 \Phi'(\varphi)^2 \left(-f'(r) + f''(r) r \right) \right)$$

Componentes independientes no nulas; $P^{abcd} = -P^{bacd} = -P^{abdc} = P^{cdab}$.

$$\left[P^{abcd} \right]^{0101} = \frac{\left(-r^2 + \beta_0 \ell^2 \Phi'(\varphi)^2 \right)}{2r^2}$$

$$\left[P^{abcd} \right]^{0202} = -\frac{\left(2r^2 + \beta_0 \ell^2 \Phi'(\varphi)^2 \right)}{4r^4 f(r)}$$

$$\left[P^{abcd} \right]^{1212} = \frac{\left(2r^2 + \beta_0 \ell^2 \Phi'(\varphi)^2 \right) f(r)}{4r^4}$$

Política de lectura. Este PDF/LaTeX es una vista. El archivo `results.json` conserva la representación intermedia canónica y la trazabilidad completa.